PROJECT HUMAN-ROBOT INTERACTION

Objective: To develop an intuitive interface between humans and the Quanser QArm robotic system using real-time computer vision and adaptive control logic.

Computer Vision

- Real-time object detection (Ball/Bottle) using Python 3.12.
- Gesture recognition for commands.
- Low-latency UDP communication.

Smart Control

- Decision-making brain in Simulink.
- Autonomous "Hunting" modes.
- Watchdog timers for fail-safe operations.

HRI Interface

- "Joystick" control via hand gestures.
- Dynamic delivery scaling.
- Human-centric collaborative logic.

Tools & Tech

- Quanser QArm Hardware.
- Inverse Kinematics & UDP protocol.
- Python & Simulink integration.

This project combines vision, control, and HRI to bridge the gap between humans and collaborative robots.